

Early-Warning Classification and Regression for Cost Overruns and Schedule Delays

A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Predictive Construction Project Overrun Model												
2	Executive early-warning view for cost overruns and schedule delays. All projects and results are synthetic.												
3													
4													
5													
6	0.899			0.774			62.5%			81.1%			
7													
8	Cost ROC-AUC			Cost PR-AUC			Cost Precision			Cost Recall			
9													
10	0.756			0.524			52.0%			45.3%			
11													
12	Schedule ROC-AUC			Schedule PR-AUC			Schedule Precision			Schedule Recall			
13													
14													
15													
16													
17													
18	Risk Band		Test Projects										
19	Red		188										
20	Yellow		138										
21	Green		110										
22													
23													
24													
25													
26													
27													
28													
29													
30													
31													
32													
33													
34													
35													
36													
37													
38													
39													
40													
41													
42													
43													
44													
45													
46													
47													
48													
49													
50													
51													
52													
53													
54													
55													
56													
57													
58													
59													
60													
61													
62													
63													
64													
65													
66													
67													
68													
69													
70													
71													
72													
73													
74													
75													
76													
77													
78													
79													
80													
81													
82													
83													
84													
85													
86													
87													
88													
89													
90													
91													
92													
93													
94													
95													
96													
97													
98													
99													
100													
101													
102													
103													
104													
105													
106													
107													
108													
109													
110													
111													
112													
113													
114													
115													
116													
117													
118													
119													
120													
121													
122													
123													
124													
125													
126													
127													
128													
129													
130													
131													
132													
133													
134													
135													
136													
137													
138													
139													
140													
141													
142													
143													
144													
145													
146													
147													
148													
149													
150													
151													
152													
153													
154													
155													
156													
157													
158													
159													
160													
161													
162													
163													
164													
165													
166													
167													
168													
169													
170													
171													
172													
173													
174													
175													
176													
177													
178													
179													
180													
181													
182													
183													
184													
185													
186													
187													
188													
189													
190													
191													
192													
193													
194													
195													
196													
197													
198													
199													
200													
201													
202													
203													
204													
205													
206													
207													
208													
209													
210													
211													
212													
213													
214													
215													
216													
217													
218													
219													
220													
221													
222													
223													
224													
225													
226													
227													
228													
229													
230													
231													
232													
233													
234													
235													
236													
237													
238													
239													
240													
241													
242													
243													
244													
245													
246													
247													
248													
249													
250													
251													
252													
253													
254													
255													
256													
257													
258													
259													
260													
261													
262													
263													
264													
265													
266													
267													
268													
269													
270													
271													
272													
273													
274													
275													
276													
277													
278													
279													
280													
281													
282													
283													
284													
285													
286													
287													
288													
289													
290													
291													
292													
293													
294													
295													
296													
297													
298													
299													
300													
301													
302													
303													
304													
305													
306													
307													
308													
309													
310													
311													
312													
313													
314													
315													
316													
317													
318													
319													
320													
321													
322													
323													
324													
325													
326													
327													
328													
329													
330													
331													
332													
333													
334													
335													
336													
337													
338													
339													
340													
341													
342													
343													
344													
345													
346													
347													
348													
349													
350													
351													
352													
353													
354													
355													
356													
357													
358													
359													
360													
361													
362													
363													
364													
365													
366													
367													
368													
369													
370													
371													
372													
373													
374													
375													
376													
377													
378													
379													
380													
381													
382													
383													
384													
385													
386													
387													
388													
389													
390													
391													
392													
393													
394													
395													
396													
397													
398													
399													
400													
401													
402													
403													
404													
405													
406													
407													
408													
409													
410													
411													
412													
413													
414													
415													
416													
417													
418													
419													
420													
421													
422													
423													
424													
425													
426													
427													
428													
429													
430													
431													
432													
433													
434													
435													
436													
437													
438													
439													
440													
441													
442													
443													
444													
445													
446													
447													
448													
449													
450													
451													
452													
453													
454													
455													
456													
457													
458													
459													
460													
461													
462													
463													
464													
465													
466													
467													
468													
469													
470													
471													
472													
473													
474													
475													
476													
477													
478													
479													
480													
481													
482													
483													
484													
485													
486													
487													
488													
489													
490													
491													
492													
493													
494													
495													
496													
497													
498													
499													
500													
501													
502													
503													
504													
505													
506													
507													
508													
509													
510													
511													
512													
513													
514													
515													
516													
517													
518													
519													
520													
521													
522													
523													
524													
525													
526													
527													
528													
529													
530													
531													
532													
533													
534													
535													
536													
537													
538													
539													
540													
541													
542													
543													
544													
545													
546													
547													
548													
549													
550													
551													
552													
553													
554													
555													
556													
557													
558													
559													
560													
561													
562													
563													
564													
565													
566													
567													
568													
569													
570													
571													
572													
573													
574													
575													
576													
577													
578													
579													
580													
581													
582													
583													
584													
585													
586													
587													
588													
589													
590													
591													
592													
593													
594													
595													
596													
597													
598													
599													
600													
601													
602													
603													
604													
605													
606													
607													
608													

Project code: IP-DA-003
Sponsor: In Project LLC
Analyst: Narciso M. Dickson, PMP®

Synthetic-data disclosure: All project records and model outcomes are fictional. Model performance is not an industry benchmark, and production use on real construction projects is not authorized by this portfolio project.

Contents

1. Executive Summary	2
2. Business Problem and Portfolio Progression	3
3. Analytical Lifecycle and Governance.....	3
4. Dataset, Feature Lineage, and Quality	4
5. Modeling Methodology	5
6. Classification Performance.....	5
7. Regression, Calibration, and Temporal Generalization	6
8. Feature Drivers and Project Risk Scoring.....	7
9. Dashboards and Portfolio Communication	8
10. Act Plan, Human Review, and Monitoring.....	10
11. Limitations, Responsible AI, and Conclusion.....	10
12. Sources and Deliverables	11

1. Executive Summary

This case study is the third project in a construction analytics portfolio. It combines the cost and schedule controls developed in Project 1 with the RFI and change-workflow indicators developed in Project 2 to predict future cost overruns and schedule delays from early and mid-project snapshots.

Dataset	Cost model	Schedule model	Test risk bands
2,362 clean projects 1,577 train 349 validation 436 test	Random Forest ROC-AUC 0.899 PR-AUC 0.774	Logistic Regression ROC-AUC 0.756 PR-AUC 0.524	188 Red 138 Yellow 110 Green

Principal conclusions

- Cost-overflow classification generalized strongly to the future synthetic test period.
- Schedule-delay performance decreased from validation to test, creating a real model-monitoring concern rather than a result to hide.
- CPI and current cost variance were the strongest cost-overflow predictors.
- SPI, planned completion, procurement delay, and overdue workflow indicators were important schedule predictors.
- Predictions were converted into Red, Yellow, and Green review priorities, but every score requires human review.
- The Act phase defines a controlled-pilot path; it does not authorize production use.

2. Business Problem and Portfolio Progression

Primary business question

Which early and mid-project indicators best predict material construction cost overruns and schedule delays, and how can management prioritize intervention before the outcomes become unavoidable?

Portfolio progression

Project	Title	Contribution
Project 1	Construction Project Controls Analytics	CPI, SPI, cost, schedule, contingency, EAC, and early-warning controls
Project 2	Construction Change Order and RFI Analytics	RFI response, change approval, pending exposure, revision loops, and forecast lag
Project 3	Predictive Construction Project Overrun Model	Time-validated probability and magnitude prediction with governance

Project 3 uses a new independent synthetic dataset. It references feature concepts from Projects 1 and 2 without mixing their folders, repositories, or records.

3. Analytical Lifecycle and Governance

Phase	CRISP-DM	DMAIC	Microsoft	PMI governance
Ask	Business Understanding	Define	Business Understanding	Initiating / Planning
Prepare	Data Understanding	Measure	Data Acquisition and Understanding	Planning

Phase	CRISP-DM	DMAIC	Microsoft	PMI governance
Process	Data Preparation	Measure / early Analyze	Data Acquisition and Understanding	Executing / Monitoring
Analyze	Modeling and Evaluation	Analyze	Modeling	Executing / Monitoring
Share	Evaluation	Analyze / improvement design	Customer Acceptance	Stakeholder engagement
Act	Deployment	Improve and Control	Deployment / Acceptance	Executing / Monitoring / Closing

The frameworks are conceptually aligned but are not treated as identical. The workflow is iterative, and quality or governance findings may require returning to earlier phases.

4. Dataset, Feature Lineage, and Quality

The deterministic generator created 2,400 intended projects with snapshots between approximately 35% and 60% planned completion. Raw tables intentionally contained duplicates, category variants, missing critical values, invalid numeric values, invalid dates, target inconsistencies, and orphan relationships.

Control	Result
Raw project rows	2,412
Raw workflow rows	2,417
Raw outcome rows	2,416
Duplicate removals	36
Quarantined records	123
Clean model rows	2,362
Cleaning actions	199
SQLite integrity	ok
Foreign-key violations	0

Feature families

- Project and contract attributes
- CPI, SPI, cost variance, schedule variance, contingency, and productivity
- Procurement, weather, and safety indicators
- RFI response, overdue rates, revisions, reopening, and RFI-to-change conversion
- Approved and pending change exposure, approval cycle, old pending rate, and forecast incorporation lag

- Workflow handoffs, revision loops, and snapshot workflow-risk score

Final outcomes, final CPI/SPI, actual end date, final EAC, and final status were excluded from predictor matrices.

5. Modeling Methodology

Element	Method
Time split	Train 2019–2023; Validation 2024; Test 2025
Cost classification	Final cost overrun $\geq 10\%$
Schedule classification	Final schedule delay ≥ 30 days
Cost regression	Final cost-overrun percentage
Schedule regression	Final schedule-delay days
Classification candidates	Logistic Regression and Random Forest
Regression candidates	Ridge Regression and Random Forest Regressor
Champion selection	Validation PR-AUC for classification; validation MAE for regression
Threshold selection	Maximum validation F1; test untouched
Explainability	Permutation importance on held-out test data

The model uses 40 predictors: 31 numeric and 9 categorical. Numeric fields are standardized for linear models, and categorical values are one-hot encoded with unknown-category handling.

6. Classification Performance

Target	Champion	ROC-AUC	PR-AUC	Precision	Recall	F1	Brier
Cost overrun	Random Forest	0.899	0.774	0.625	0.811	0.706	0.123
Schedule delay	Logistic Regression	0.756	0.524	0.520	0.453	0.484	0.183

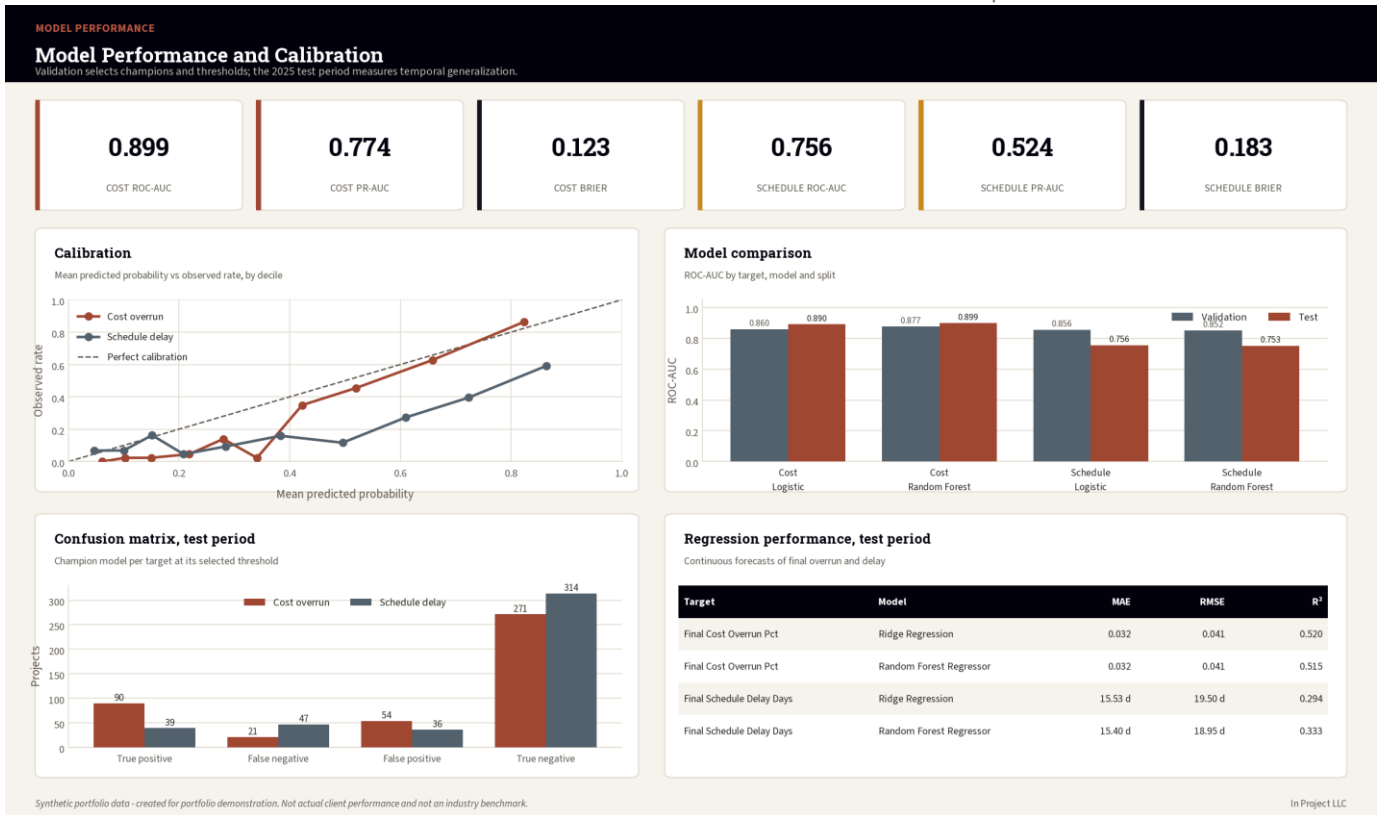


Figure 1. Model performance and calibration: champion selection on validation, temporal generalization measured on the 2025 test period.

7. Regression, Calibration, and Temporal Generalization

Target	Model	MAE	RMSE	R ²	Interpretation
Final_Cost_Overrun_Pct	Ridge Regression	0.032	0.041	0.520	Cost-overrun percentage points
Final_Cost_Overrun_Pct	Random Forest Regressor	0.032	0.041	0.515	Cost-overrun percentage points
Final_Schedule_Delay_Days	Ridge Regression	15.52 8	19.49 9	0.294	Schedule-delay days
Final_Schedule_Delay_Days	Random Forest Regressor	15.40 3	18.95 3	0.333	Schedule-delay days

Temporal degradation

The schedule classifier achieved validation ROC-AUC 0.856 but test ROC-AUC 0.756. The reduction suggests temporal variation in the synthetic portfolio. The result is retained as an example of why model performance must be monitored after deployment.

The monitoring demonstration produced a schedule probability PSI of 0.105, above the proposed Yellow threshold of 0.10.

8. Feature Drivers and Project Risk Scoring

Rank	Cost feature	Importance	Schedule feature
1	CPI_Snapshot	0.0502	SPI_Snapshot
2	Cost_Variance_Pct_Snapshot	0.0289	Planned_Percent_Complete
3	Approved_Change_Value_Pct_Budget	0.0080	Procurement_Delay_Days_to_Date
4	Pending_Change_Exposure_Pct_Budget	0.0030	Overdue_RFI_Rate
5	Labor_Productivity_Index	0.0026	Old_Pending_Change_Rate
6	Contingency_Utilization_Pct	0.0022	Actual_Percent_Complete
7	Procurement_Delay_Days_to_Date	0.0020	RFI_On_Time_Rate
8	SPI_Snapshot	0.0013	Revision_Loop_Rate
9	Snapshot_Data_Completeness_Pct	0.0012	Complexity_Level
10	Complexity_Level	0.0010	Project_Type

Feature importance measures predictive contribution under permutation. It does not establish that changing a feature will cause a better outcome.

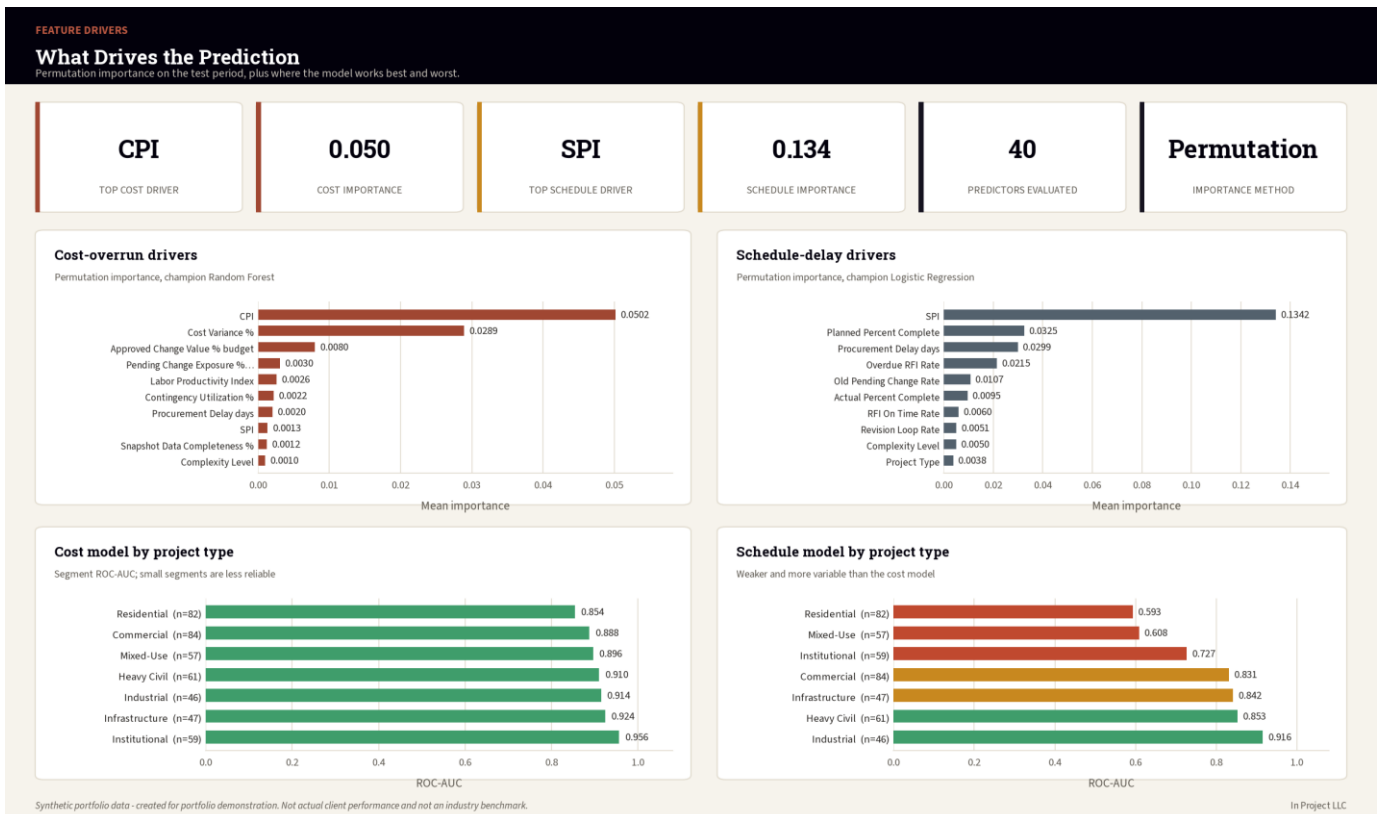


Figure 2. Permutation importance for both targets, with segment performance showing where each model is weakest.

Top predicted-risk projects

Project	Band	Overall	Cost	Schedule	Primary drivers
Gateway Bridge Program 1418	Red	99.8%	94.0%	97.4%	Slow change approval; Workflow risk score; Overdue RFI backlog
Granite Urban Village 1705	Red	99.8%	89.1%	97.7%	Overdue RFI backlog; Slow change approval; Workflow risk score
Horizon Neighborhood 34	Red	99.4%	80.3%	96.8%	Overdue RFI backlog; Workflow risk score; Slow RFI response
Redwood Logistics Center 1716	Red	99.2%	91.6%	90.9%	Overdue RFI backlog; Slow RFI response; Workflow risk score
Aspen Corporate Center 702	Red	99.1%	91.8%	89.2%	Overdue RFI backlog; Slow change approval; Slow RFI response
Northstar Highway Improvements 1925	Red	99.0%	88.1%	92.0%	Slow change approval; Workflow risk score; Slow RFI response
Pioneer Urban Village 1981	Red	98.8%	94.1%	80.4%	Overdue RFI backlog; Slow RFI response; Workflow risk score

9. Dashboards and Portfolio Communication

The Share phase produced executive, model-performance, project-risk, and feature-driver dashboards plus Power BI and Tableau build specifications.

EXECUTIVE MODEL VIEW

Predictive Overrun Model

Early-warning classification and regression across 2,362 synthetic construction projects, validated on a future 2025 test period.

2,362

MODELING PROJECTS

40

PREDICTORS

0.899

COST ROC-AUC (TEST)

0.756

SCHEDULE ROC-AUC (TEST)

188

RED-BAND PROJECTS

436

TEST PROJECTS SCORED

Predicted risk bands

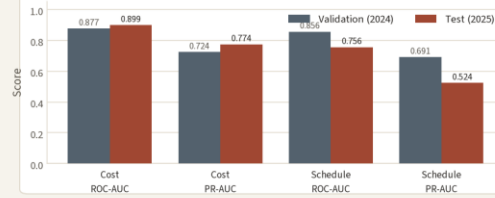
2025 test period, combined probability



Red 188 (43%)
Yellow 138 (32%)
Green 110 (25%)

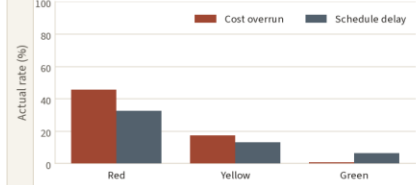
Champion model performance

Validation selects the model; test measures generalization



Outcome rate by predicted band

Actual overrun and delay rate in each band



Highest predicted risk projects

Ranked by combined overrun probability; every row requires human review

#	Project	Type	Region	CPI	SPI	Cost prob	Sched prob	Combined	Band	Primary drivers
1	PRJ3-01418 Gateway Bridge Program...	Heavy Civil	West	0.881	0.845	94.0%	97.4%	99.8%	Red	Slow change approval; Workflow...
2	PRJ3-01705 Granite Urban Village...	Mixed-Use	Midwest	0.909	0.824	89.1%	97.7%	99.8%	Red	Overdue RFI backlog; Slow change...
3	PRJ3-00034 Horizon Neighborhood 34	Residential	Southeast	0.912	0.922	80.3%	96.8%	99.4%	Red	Overdue RFI backlog; Workflow...
4	PRJ3-01716 Redwood Logistics...	Industrial	West	0.864	0.891	91.6%	90.9%	99.2%	Red	Overdue RFI backlog; Slow RFI...
5	PRJ3-00702 Aspen Corporate Center...	Commercial	Northeast	0.861	0.907	91.8%	89.2%	99.1%	Red	Overdue RFI backlog; Slow change...
6	PRJ3-01925 Northstar Highway...	Heavy Civil	Midwest	0.818	0.904	88.1%	92.0%	99.0%	Red	Slow change approval; Workflow...
7	PRJ3-01981 Pioneer Urban Village...	Mixed-Use	Mountain	0.839	0.965	94.1%	80.4%	98.8%	Red	Overdue RFI backlog; Slow RFI...
8	PRJ3-01492 Horizon Education...	Institutional	Southwest	0.823	0.859	92.1%	82.4%	98.6%	Red	Overdue RFI backlog; Low CPI; Low...
9	PRJ3-01752 Lakeview Highway...	Heavy Civil	West	0.800	0.860	87.1%	88.3%	98.5%	Red	Overdue RFI backlog; Low CPI; Low...
10	PRJ3-02338 Riverside Neighborhood...	Residential	Mountain	0.937	0.952	74.5%	92.9%	98.2%	Red	Overdue RFI backlog; Slow change...

Synthetic portfolio data - created for portfolio demonstration. Not actual client performance and not an industry benchmark.

In Project LLC

Figure 3. Executive model view: predicted risk bands, champion performance and the highest predicted-risk projects.

PROJECT RISK

Project Risk Prioritization

Which projects to review first, and how the two risks combine.

188

RED BAND

138

YELLOW BAND

110

GREEN BAND

25.5%

ACTUAL COST-OVERRUN RATE

19.7%

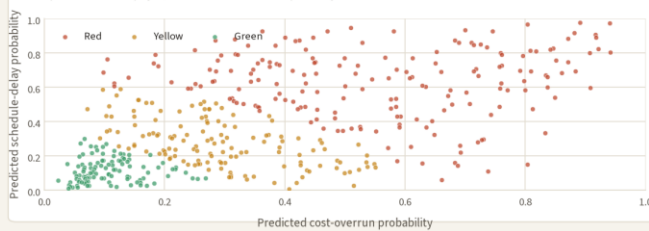
ACTUAL DELAY RATE

Human review

REQUIRED BEFORE ACTION

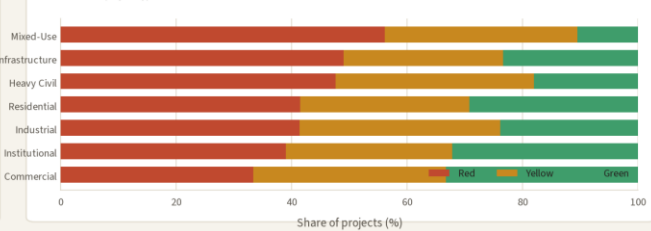
How the two risks combine

Each point is a 2025 test project; band comes from combined probability



Risk band by project type

Share of each project type in each band



Top 12 projects for management review

Highest combined probability; predictions are decision support, not decisions

#	Project	Type	Delivery	CPI	SPI	Overdue RFI	Approval days	Cost prob	Sched prob	Combined	Band
1	PRJ3-01418 Gateway Bridge...	Heavy Civil	Design-Bid-Build	0.881	0.845	46.1%	45.6	94.0%	97.4%	99.8%	Red
2	PRJ3-01705 Granite Urban Village...	Mixed-Use	CMAR	0.909	0.824	52.2%	37.6	89.1%	97.7%	99.8%	Red
3	PRJ3-00034 Horizon Neighborhood...	Residential	Design-Bid-Build	0.912	0.922	59.5%	35.6	80.3%	96.8%	99.4%	Red
4	PRJ3-01716 Redwood Logistics...	Industrial	Design-Bid-Build	0.864	0.891	51.7%	30.9	91.6%	90.9%	99.2%	Red
5	PRJ3-00702 Aspen Corporate...	Commercial	Design-Bid-Build	0.861	0.907	48.9%	34.6	91.8%	89.2%	99.1%	Red
6	PRJ3-01925 Northstar Highway...	Heavy Civil	Design-Bid-Build	0.818	0.904	42.2%	38.5	88.1%	92.0%	99.0%	Red
7	PRJ3-01981 Pioneer Urban Village...	Mixed-Use	CMAR	0.839	0.965	56.9%	31.2	94.1%	80.4%	98.8%	Red
8	PRJ3-01492 Horizon Education...	Institutional	Design-Bid-Build	0.823	0.859	42.4%	23.0	92.1%	82.4%	98.6%	Red
9	PRJ3-01752 Lakeview Highway...	Heavy Civil	Design-Bid-Build	0.800	0.860	46.3%	25.8	87.1%	88.3%	98.5%	Red
10	PRJ3-02338 Riverside...	Residential	Design-Bid-Build	0.937	0.952	54.3%	34.8	74.5%	92.9%	98.2%	Red
11	PRJ3-00088 Harbor District 88	Mixed-Use	Design-Bid-Build	0.839	0.888	55.8%	27.5	90.7%	80.2%	98.2%	Red
12	PRJ3-01155 Copper Education...	Institutional	CMAR	0.899	0.840	40.8%	33.6	70.0%	93.4%	98.0%	Red

Synthetic portfolio data - created for portfolio demonstration. Not actual client performance and not an industry benchmark.

In Project LLC

Figure 4. Project risk prioritization: how cost and schedule probability combine into a single review queue.

The communication design shows probability and risk band together, labels validation and test periods, and keeps synthetic-data, uncertainty, and human-review warnings visible.

10. Act Plan, Human Review, and Monitoring

Controlled-pilot roadmap

- Days 0–30: approve charter, prohibited uses, independent review, reproducibility, source mapping, and data-quality gates.
- Days 31–60: conduct a silent pilot, train reviewers, and review thresholds and monitoring.
- Days 61–90: conduct a limited visible pilot, record decisions and overrides, and make a formal go/no-go decision.

Human review

1. Verify input data and snapshot context.
2. Review cost, schedule, and overall probabilities.
3. Review model and rule-based drivers.
4. Consider current project context and known events.
5. Accept, Override, Escalate, or Defer.
6. Assign an action, owner, due date, and evidence.
7. Record later outcomes for monitoring.

Proposed monitoring

Metric	Yellow	Red / stop
Cost ROC-AUC	Yellow <0.82	Red <0.78
Schedule ROC-AUC	Yellow <0.75	Red <0.70
Probability PSI	Yellow >0.10	Red >0.20
Critical data quality	Any failure	Stop scoring
Segment AUC gap	Yellow >0.10	Red >0.15
Override rate	Yellow >30%	Red >50%
Model age	Review at 12 months	Mandatory decision at 18 months

11. Limitations, Responsible AI, and Conclusion

- All data is synthetic and may not represent real project distributions.
- Performance is not an industry benchmark.
- Target thresholds are case-study definitions.

- Feature importance is predictive, not causal.
- The schedule model shows temporal degradation.
- Segment results have limited sample sizes.
- Real deployment requires approved data, independent validation, calibration, drift review, human oversight, access controls, and incident response.
- The model may not be used for contractual, legal, safety-critical, or personnel decisions.

Conclusion

Project 3 completes a three-project construction analytics portfolio progression from descriptive project controls, to diagnostic workflow analytics, to predictive early warning. The project demonstrates technical modeling and business governance together, including the discipline to retain unfavorable temporal results and restrict unvalidated production use.

12. Sources and Deliverables

Professional references

IBM CRISP-DM overview: <https://www.ibm.com/docs/en/spss-modeler/18.6.0?topic=dm-crisp-help-overview>

ASQ DMAIC: <https://asq.org/quality-resources/dmaic>

Microsoft Data Science Lifecycle: <https://learn.microsoft.com/en-us/shows/dev-intro-to-data-science/what-is-the-data-science-lifecycle-2-of-28>

PMI Process Groups: <https://www.pmi.org/standards/process-groups>

Primary project deliverables

- Analysis workbook and model-performance tables
- Saved classification and regression pipelines
- Feature manifest and model card
- Executive and operational dashboards
- Power BI and Tableau specifications
- Batch scoring and model-monitoring scripts
- Controlled-pilot Act plan, risk register, RACI, and override log
- Complete Ask–Prepare–Process–Analyze–Share–Act documentation

End of Report